

Suppose a family of pdf/pmf $\{f(x|\theta): \theta \in \Theta \subseteq \mathbb{R}\}$
has MLR in $T(x)$ if $\theta_2 > \theta_1$

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Para Inf.

i.e. $\frac{f(x|\theta_2)}{f(x|\theta_1)}$ non-decreasing in $T(x) \quad \forall x \in \{x: f(x|\theta_1) > 0\} \cup \{x: f(x|\theta_2) > 0\}$

Karlin-Rubin: Consider testing $H_0: \theta \leq \theta_0$ vs. $H_1: \theta > \theta_0 \dots (*)$

(i) For fixed $0 < \alpha < 1 \quad \exists$ UMP size- α test for $(*)$ given by

$$\phi^*(x) = \begin{cases} 1 & \text{if } T(x) > c \\ \gamma & \text{if } T(x) = c \\ 0 & \text{if } T(x) < c \end{cases} \quad \text{where } c \in \mathbb{R} \text{ and } \gamma \in [0, 1] \text{ is determined from } \beta_{\phi^*}(\theta_0) = \mathbb{E}_{\theta_0} \phi^* = \alpha$$

(ii) The power fn. $\beta_{\phi^*}(\theta)$ is strictly increasing for θ for which $0 < \beta_{\phi^*}(\theta) < 1$

(iii) For any $\theta < \theta_0$, ϕ^* minimizes $\beta_{\phi}(\theta)$ among all test ϕ that satisfy $\beta_{\phi}(\theta_0) = \alpha$
 $\underbrace{\beta_{\phi}(\theta_0)}_{\mathbb{P}(\text{Type I error})}$

i.e. $\beta_{\phi^*}(\theta) \leq \beta_{\phi}(\theta)$

Pf: (i) Fix $\theta_1 > \theta_0$. Consider testing $H_0': \theta = \theta_0$ vs. $H_1': \theta = \theta_1$

Note: $\frac{f(x|\theta_1)}{f(x|\theta_0)} = \frac{h(x) g_{\theta_1}(T(x))}{h(x) g_{\theta_0}(T(x))} = \psi_{\theta_1, \theta_0}(T(x))$ for some non-decreasing for ψ of $T(x)$ ($\because$ MLR)

Existence:

Def: $h(t) := \mathbb{P}_{\theta_0}(T(X) > t) \downarrow t$ (non-increasing)

$$= 1 - F_T(t)$$

$$h(\infty) = 0, \quad h(-\infty) = 1$$

$\therefore \exists c \in \mathbb{R} \quad \underbrace{h(c) \leq \alpha \leq h(c-)}_{\text{Show size} = \alpha} \quad \text{Set } \gamma = \begin{cases} \frac{\alpha - h(c)}{h(c-) - h(c)} & \text{if } h(c) \neq h(c-) \\ 0 & \text{if } = \end{cases}$

Sufficiency: T.S. ϕ^* is MP size- α for H_0' vs. H_1'

$$\beta_{\phi^*}(\theta_1) \geq \beta_{\phi}(\theta_1) \text{ for all } \phi \text{ satisfying } \beta_{\phi}(\theta_0) = \alpha \quad (\leq \alpha)$$

$$A = \{x: \phi^*(x) \neq \phi(x)\} \quad \text{If } x \in A, \phi^*(x) - \phi(x) > 0$$

$$\Rightarrow T(x) \geq c \quad (\text{b/c if } T(x) > c \Rightarrow \phi^* = 0, \phi' > 0 \Rightarrow \phi^* - \phi < 0)$$

$$\Rightarrow \psi(T(x)) \geq \psi(c) = K \Rightarrow f(x|\theta_1) \geq K f(x|\theta_0)$$

$$\text{If } x \in A, \phi^* > \phi \Rightarrow f(x|\theta_1) \geq K f(x|\theta_0)$$

$$\text{If } x \in A, \phi^* < \phi \Rightarrow f(x|\theta_1) \leq K f(x|\theta_0)$$

$$g(x) = (\phi^*(x) - \phi(x))(f(x|\theta_1) - K f(x|\theta_0)) \geq 0 \quad \forall x \in A$$

$$\int_{\mathbb{R}^n} (\phi^*(x) - \phi(x))(f(x|\theta_1) - K f(x|\theta_0)) dx \geq 0$$

$$\Rightarrow \mathbb{E}_{\theta_1} \phi^* - \mathbb{E}_{\theta_1} \phi \geq K \left(\underbrace{\mathbb{E}_{\theta_0} \phi^*}_{\alpha} - \underbrace{\mathbb{E}_{\theta_0} \phi}_{\leq \alpha} \right) \geq 0$$

$$\Rightarrow \beta_{\phi^*}(\theta_1) \geq \beta_{\phi}(\theta_1)$$

i.e. ϕ^* is UMP test for $H_0': \theta = \theta_0$ vs. $H_1'': \theta > \theta_0$ since the ϕ^* does not depend on the choice of θ_1 (why?)

$$\beta_{\phi^*}(\theta_0) = \alpha$$

$$\mathbb{P}_{\theta_0}(T(x) > c) + \gamma \mathbb{P}_{\theta_0}(T(x) = c) = \alpha$$

T.S.

$$\beta_{\phi^*}(\theta) \leq \beta_{\phi^*}(\theta_0) \quad \forall \theta \leq \theta_0$$

Recall Lemma

If $\{f_\theta(x) : \theta \in \Theta \subseteq \mathbb{R}\}$ has MLR in $T(x)$
 ψ is a non-decreasing fn. of $T(x)$

then $g(\theta) := E_\theta(\psi(T(x)))$ is non-decreasing in θ

$$\phi^*(x) = \mathbb{1}(T(x) > c) + \gamma \mathbb{1}(T(x) = c)$$

$$= \psi(T(x)) \text{ is non-decreasing in } T(x)$$

$\therefore$ By Lemma

$$\beta_{\phi^*}(\theta) = E_\theta[\phi^*(x)] \text{ is } \uparrow \theta$$

$$\Rightarrow \beta_{\phi^*}(\theta) \leq \beta_{\phi^*}(\theta_0) \quad \forall \theta \leq \theta_0$$

$$\Rightarrow \sup_{\theta \leq \theta_0} \beta_{\phi^*}(\theta) = \beta_{\phi^*}(\theta_0) = \alpha$$

$\phi^* = 0$	$\phi^* = \gamma$	$\phi^* = 1$
$T(x) < c$	c	$T(x) > c$

(ii) $0 < \beta_{\phi^*}(\theta) < 1$ consider $\theta_1 < \theta_2$

$$\beta_{\phi^*}(\theta_1) \leq \beta_{\phi^*}(\theta_2)$$

Step 1: Consider testing $H_0': \theta = \theta_1$ vs. $H_1': \theta = \theta_2$
 To show ϕ^* is a MP test for H_0' vs. H_1'
 with size $\alpha_1 := \beta_{\phi^*}(\theta_1)$

T.S. $\beta_{\phi^*}(\theta_2) \geq \beta_{\phi}(\theta_2)$ for any ϕ satisfying $\beta_{\phi}(\theta_1) = \alpha_1$

$$A = \{x : \phi^*(x) \neq \phi(x)\}$$

$$\text{If } x \in A, \phi^* > \phi \Rightarrow T(x) \geq c$$

$$\left[\frac{f(x|\theta_2)}{f(x|\theta_1)} = \psi_{\theta_1, \theta_2}(T(x)) \text{ for some non-decreasing fn } \psi \right]$$

$$\Rightarrow \psi(T(x)) \geq \psi(c) = K \text{ (say)}$$

$$\Rightarrow f(x|\theta_2) \geq K f(x|\theta_1)$$

$$\text{If } x \in A, \phi^* < \phi \Rightarrow f(x|\theta_2) \leq K f(x|\theta_1)$$

$$\int (\phi^*(x) - \phi(x)) (f(x|\theta_2) - K f(x|\theta_1)) dx \geq 0$$

$$\Rightarrow E_{\theta_2} \phi^* - E_{\theta_2} \phi \geq K[\alpha_1 - \alpha_1] = 0$$

If $0 < \beta_{\phi^*}(\theta_1) = \beta_{\phi^*}(\theta_2) < 1$ then ϕ^* is MP size- α_1 test with power β

Consider test $\phi_0(x) = \alpha_1 \forall x$ then ϕ_0 is also MP size- α_1 test for H_0' vs. H_1'

By uniqueness $\phi_0 = \begin{cases} 1 & \text{if } f(x|\theta_2) > K f(x|\theta_1) \\ 0 & \text{if } f(x|\theta_2) < K f(x|\theta_1) \end{cases}$ But $\phi_0 \in (0,1)$

$$\Rightarrow f(x|\theta_2) = K f(x|\theta_1)$$

$$\Rightarrow K = 1$$

$$\Rightarrow f(x|\theta_2) = f(x|\theta_1) \text{ a.e.}$$

(iii) For any $\theta < \theta_0$

$$\beta_{\phi^*}(\theta) \leq \beta_{\phi}(\theta) \text{ for any test } \phi: \beta_{\phi}(\theta_0) = \beta_{\phi^*}(\theta_0) = \alpha$$

Fix $\theta' < \theta_0$ Consider testing $H_0: \theta = \theta_0$ vs. $H_1: \theta = \theta'$

Note: $\frac{f(x|\theta')}{f(x|\theta_0)} = \psi_{\theta_0, \theta'}(T(x))$ for some non-increasing ψ in $T(x)$

$$A = \{x: \phi^*(x) \neq \phi(x)\}$$

$$\text{If } x \in A, \phi^* > \phi \Rightarrow T(x) \geq c$$

$$\Rightarrow \psi(T(x)) \leq \psi(c) = K$$

$$\Rightarrow f(x|\theta') \leq K f(x|\theta_0)$$

$$\text{If } x \in A, \phi^* > \phi \Rightarrow T(x) \leq c$$

$$\Rightarrow f(x|\theta') \leq K f(x|\theta_0)$$

$$\Rightarrow (\phi^*(x) - \phi(x))(f(x|\theta') - K f(x|\theta_0)) \leq 0 \quad \forall x \in A$$

$$\Rightarrow \mathbb{E}_{\theta'} \phi^* - \mathbb{E}_{\theta'} \phi \leq 0$$

$$\Rightarrow \beta_{\phi^*}(\theta') - \beta_{\phi}(\theta') \leq 0$$

$$\because \theta' \text{ arbitrary, } \beta_{\phi^*}(\theta) \leq \beta_{\phi}(\theta) \quad \forall \theta < \theta_0$$

Karlin-Rubin

(iv) Assume $\mathbb{P}_{\theta}(f(x|\theta) = c f(x|\theta_0)) = 0 \quad \forall \theta > \theta_0$ and for any $c \geq 0$. If a test ϕ satisfies $\beta_{\phi}(\theta_0) = \beta_{\phi^*}(\theta_0) (= \alpha)$, then for any $\theta > 0$ either $\beta_{\phi^*}(\theta) > \beta_{\phi}(\theta)$ or $\phi^*(x) = \phi(x)$ $\mathbb{P}_{\theta}$ -a.e. ($\mathbb{P}_{\theta}(\phi^*(x) = \phi(x)) = 1$)

Pl: Fix $\theta_1 > \theta_0$, consider testing $H'_0: \theta = \theta_0$ vs. $H'_1: \theta = \theta_1$

Know ϕ^* is MP size- α test for H'_0 vs. H'_1 (from part (i))

For H'_0 vs. H'_1 , ϕ is a size- α test

$$\Rightarrow \beta_{\phi^*}(\theta_1) \geq \beta_{\phi}(\theta_1)$$

Suppose $\beta_{\phi^*}(\theta_1) = \beta_{\phi}(\theta_1) \Rightarrow \phi$ is MP size- α test for H_0' vs. H_1'

$$(\phi^*(x) - \phi(x))(f(x|\theta_1) - Kf(x|\theta_0)) = 0 \quad \mu. a.e.$$

(Recall uniqueness of NP lemma)

$$\phi^*(x) = \phi(x) \text{ or } f(x|\theta_1) = Kf(x|\theta_0) \quad \mu. a.e.$$

$$P_{\theta_i}(\underbrace{\{\phi^*(x) = \phi(x)\}}_{B_1} \cup \underbrace{\{f(x|\theta_1) = Kf(x|\theta_0)\}}_{B_2}) = 1 \quad i=0,1$$

$$P_{\theta_1}(\{\phi^*(x) = \phi(x)\} \cup \{f(x|\theta_1) = Kf(x|\theta_0)\}) = 1$$

$$P_{\theta_1}(\phi^*(x) = \phi(x)) = 1$$

$$\begin{aligned} \text{(Given } P_{\theta_1}(B_2) = 0 \Rightarrow P(B_1 \cap B_2) = 0 \\ \Rightarrow P(B_1 \cup B_2) = P(B_1) + P(B_2) - P(B_1 \cap B_2)) \end{aligned}$$

Karlin-Rubin

(*) Fix $\theta_1 \in \Theta$. The test ϕ^* is UMP for $H_0': \theta \leq \theta_1$ vs. $H_1': \theta > \theta_1$ with size $= \beta_{\phi^*}(\theta_1)$

$\theta \geq \theta_0$ vs. $\theta < \theta_0$

Suppose $\{f(x|\theta): \theta \in \Theta \subseteq \mathbb{R}\}$ has MLR in $T(x)$. Test: $H_0: \theta \geq \theta_0$ vs. $H_1: \theta < \theta_0$

(i) For fixed $0 < \alpha < 1$ $\exists$ UMP size- α test for (*) given by

$$\phi^*(x) = \begin{cases} 1 & \text{if } T(x) < c \\ \gamma & \text{if } T(x) = c \\ 0 & \text{if } T(x) > c \end{cases} \quad \text{where } c \in \mathbb{R} \text{ and } \gamma \in [0,1] \text{ is determined from } \beta_{\phi^*}(\theta_0) = E_{\theta_0} \phi^* = \alpha$$

(ii) The power fn. $\beta_{\phi^*}(\theta)$ is strictly decreasing for θ for which $0 < \beta_{\phi^*}(\theta) < 1$

Ex (1). $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma_0^2)$. Find UMP size- α test
 $\hookrightarrow$ known for $H_0: \mu \geq \mu_0$ vs. $H_1: \mu < \mu_0$

$$\phi^* = \mathbb{1}(\bar{X} < c)$$

$$\Rightarrow \phi^*(\underline{X}) = \left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} < z_{1-\alpha} \right)$$

(2) $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu_0, \sigma^2)$ UMP size- α test for
 $H_0: \sigma \geq \sigma_0$ vs. $H_1: \sigma < \sigma_0$

$$\phi^* = \mathbb{1} \left(\frac{\sum (x_i - \mu_0)^2}{\sigma_0^2} < \chi_{n, 1-\alpha}^2 \right) \text{ is the UMP size } \alpha\text{-test}$$

for H_0 vs. H_1

$$\text{Power fn. } P_\sigma(\chi_n^2 < \frac{\sigma_0^2}{\sigma^2} \chi_{n, 1-\alpha}^2) = F_{\chi_n^2} \left(\frac{\sigma_0^2}{\sigma^2} \chi_{n, 1-\alpha}^2 \right) \downarrow \sigma$$

(3) $X_1, \dots, X_n \stackrel{iid}{\sim} U(-\theta, \theta)$. Find UMP test for $H_0: \theta \geq \theta_0$ vs.
 $H_1: \theta < \theta_0$

$$Y_i = |X_i| \quad Y_1, \dots, Y_n \stackrel{iid}{\sim} U(0, \theta)$$

$$f(y|\theta) = \frac{1}{\theta^n} \mathbb{1}(0 \leq y_{(n)} \leq \theta) \text{ has MLR in } y_{(n)}$$

$$\therefore \text{By Karlin-Rubin } \phi^* = \mathbb{1}(y_{(n)} < c)$$